

Computing upper and lower bounds on likelihoods in intractable networks

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Problem statement & proposed solution

Problem. In sigmoid and noisy-OR belief networks, exact likelihoods of partially observed variables can be intractable due to large cliques / exponential summations.

Goal. Compute *rigorous* upper and lower bounds on such likelihoods to form confidence intervals and enable decision-making without exact inference.

Proposed solution.

- ▶ **Variational upper bounds** that factorize the intractable sum by introducing *variational parameters* and exchanging minimization with marginalization (two-level networks).
- ▶ **Mean-field lower bounds** via Jensen's inequality with tractable factorized Q over unobserved nodes (generic networks).

Networks covered. Sigmoid belief networks and noisy-OR networks.

Theoretic introduction

Graphical models. DAG-structured joint $P(\mathbf{S} \mid \theta) = \prod_i P(S_i \mid \text{pa}[i], \theta)$; exact inference by junction trees may be infeasible when cliques are large.

- ▶ **Sigmoid network (binary S_i):**

$$P(S_i = 1 \mid \text{pa}[i]) = g\left(\sum_{j \in \text{pa}[i]} \theta_{ij} S_j\right) \text{ with } g(x) = \frac{1}{1+e^{-x}}.$$

- ▶ **Noisy-OR:**

$$P(S_i = 0 \mid \text{pa}[i]) = \prod_{j \in \text{pa}[i]} (1 - q_{ij})^{S_j} = \exp\left(-\sum_{j \in \text{pa}[i]} \lambda_{ij} S_j\right) \\ \text{with } \lambda_{ij} = -\log(1 - q_{ij}) \geq 0.$$

Why bounds? Exact confidence intervals for probabilities in general BNs are NP-hard; efficient *bounds* can still be tight enough for practice.

Upper bound for sigmoid network (two-level)

Setup. Bipartite $L_2 \rightarrow L_1$, observe all $\{S_i\}_{i \in L_1}$, hidden $\{S_j\}_{j \in L_2}$; write $y_i = 2S_i - 1 \in \{-1, +1\}$ and let $\pi_j := P(S_j = 1)$.

Sketch.

1. Insert the variational transform for the logistic:

$$g(x) = \frac{1}{1 + e^{-x}} = \min_{\xi \in [0,1]} \exp(-H(\xi) + \xi x),$$

with binary entropy $H(\xi) = -\xi \log \xi - (1 - \xi) \log(1 - \xi)$.

2. Apply to each observed node $g(y_i \sum_j \theta_{ij} S_j)$, collect terms by each hidden S_j ; the augmented joint factorizes over L_2 .
3. *Exchange minimization with marginalization* and sum out S_j in closed form; finally minimize over ξ .

Upper bound for sigmoid network (two-level)

Closed-form upper bound.

$$P(\{S_i\}_{i \in L_1} \mid \theta) \leq \min_{\xi \in [0,1]^{|L_1|}} \exp\left(-\sum_{i \in L_1} H(\xi_i)\right) \times \prod_{j \in L_2} \left[\pi_j \exp\left(\sum_{i \in L_1} \xi_i y_i \theta_{ij}\right) + (1 - \pi_j) \right].$$

Log form (for convex optimization). Using

$$\log x = \min_{\alpha > 0} \{\alpha x - \log \alpha - 1\},$$

$$\log P \leq -\sum_{i \in L_1} H(\xi_i) + \sum_{j \in L_2} \log \left[\pi_j e^{\sum_i \xi_i y_i \theta_{ij}} + (1 - \pi_j) \right]$$

or equivalently with auxiliaries $\alpha_j > 0$,

$$\log P \leq -\sum_i H(\xi_i) + \sum_j \alpha_j \left[\pi_j e^{\sum_i \xi_i y_i \theta_{ij}} + (1 - \pi_j) \right] - \sum_j (\log \alpha_j + 1).$$

Lower bound for sigmoid network (generic)

Setup. Evidence on L_1 with $y_i = 2S_i - 1$. Introduce a mean-field posterior over the hidden layer:

$$Q(\mathbf{S}_{L_2}) = \prod_{j \in L_2} \mu_j^{S_j} (1 - \mu_j)^{1-S_j} \approx P(\mathbf{S}_{L_2} \mid \mathbf{S}_{L_1}).$$

Sketch.

1. Apply Jensen's inequality to the intractable sum over uninstantiated variables:

$$\log P(\mathbf{S}_{L_1}) \geq \mathbb{E}_Q[\log P(\mathbf{S}_{L_1}, \mathbf{S}_{L_2})] + H(Q).$$

2. Decompose the joint under the sigmoid BN:

$$\mathbb{E}_Q[\log P(\mathbf{S}_{L_1}, \mathbf{S}_{L_2})] = \sum_{i \in L_1} \mathbb{E}_Q[\log P(S_i \mid \text{pa}[i])] + \sum_{j \in L_2} \mathbb{E}_Q[\log P(S_j)].$$

3. Under factorized Q , these expectations depend only on the means μ_j ; adjust $\{\mu_j\}$ to tighten the bound.

Lower bound for sigmoid network (generic)

Mean-field lower bound.

$$\log P(\{S_i\}_{i \in L_1} \mid \theta) \geq \sum_{i \in L_1} \mathbb{E}_Q[\log P(S_i \mid \text{pa}[i]; \theta)] + \sum_{j \in L_2} \mathbb{E}_Q[\log P(S_j)] + H(Q)$$

Notes (optimization).

- ▶ Maximize the RHS over μ ; the bound is exact if Q matches the true posterior.
- ▶ The accuracy of the bound is governed by $\text{KL}(Q \parallel P(\mathbf{S}_{L_2} \mid \mathbf{S}_{L_1}))$.

Upper bound for noisy-OR network (two-level)

Setup. Bipartite $L_2 \rightarrow L_1$. For child i (no leak, as in the paper):

$$P(S_i = 1 \mid \mathbf{S}_{L_2}) = 1 - \exp\left(-\sum_j \lambda_{ij} S_j\right),$$

$$P(S_i = 0 \mid \mathbf{S}_{L_2}) = \exp\left(-\sum_j \lambda_{ij} S_j\right),$$

with $\lambda_{ij} = -\log(1 - q_{ij}) \geq 0$ and $\pi_j := P(S_j = 1)$.

Sketch.

1. Use the variational identity

$$1 - e^{-x} = \min_{\eta \geq 0} \exp(-\eta x - F(\eta)), \quad F(\eta) = -\eta \log \eta + (\eta + 1) \log(\eta + 1).$$

2. Insert per observed node, regroup by parents (L_2): the augmented joint factorizes over S_j .
3. Exchange the sum over $\mathbf{S}_{L_2}$ with the variational step to obtain a closed form in $\{\eta_i\}$, then (optionally) optimize them.

Upper bound for noisy-OR network (two-level)

Closed-form upper bound.

$$P(\{S_i\}_{i \in L_1}) \leq \min_{\{\eta_i \geq 0\}} \left\{ \exp\left(-\sum_{i \in L_1} S_i F(\eta_i)\right) \times \prod_{j \in L_2} \left[\pi_j \exp\left(-\sum_{i \in L_1} (S_i \eta_i + 1 - S_i) \lambda_{ij}\right) + (1 - \pi_j) \right] \right\}.$$

Log form. Using $\log x = \min_{\alpha > 0} \{\alpha x - \log \alpha - 1\}$ and introducing auxiliaries $\{\alpha_j > 0\}$,

$$\log P \leq -\sum_{i \in L_1} S_i F(\eta_i) + \sum_{j \in L_2} \alpha_j \left[\pi_j \exp\left(-\sum_{i \in L_1} (S_i \eta_i + 1 - S_i) \lambda_{ij}\right) + (1 - \pi_j) \right] - \sum_{j \in L_2} (\log \alpha_j + 1).$$

Lower bound for noisy-OR network (generic)

Setup. Reuse the same factorized Q over the hidden parents and apply Jensen to $\log P(\mathbf{S}_{L_1})$ as before.

Sketch.

1. Use the expansion $1 - e^{-x} = \prod_k g(2^k x)$ with $g(u) = 1/(1 + e^{-u})$.
2. Take logs and apply Jensen under Q to the intractable terms.
3. Use convexity to move expectations inside $\log(1 + \cdot)$.
4. Compute $\mathbb{E}_Q \left[e^{-2^k \sum_j \lambda_{ij} S_j} \right] = \prod_j (\mu_j e^{-2^k \lambda_{ij}} + (1 - \mu_j))$; for $S_i = 0$ the term is exact: $-\sum_j \lambda_{ij} \mu_j$.

Lower bound for noisy-OR network (generic)

Closed-form.

$$P(\{S_i\}_{i \in L_1}) \geq e^{H(Q)} \exp\left(- \sum_{i: S_i=0} \sum_{j \in L_2} \lambda_{ij} \mu_j\right) \\ \prod_{i: S_i=1} \prod_{k=0}^{\infty} \left(1 + \prod_{j \in L_2} ((1 - \mu_j) + \mu_j e^{-2^k \lambda_{ij}})\right)^{-1}.$$

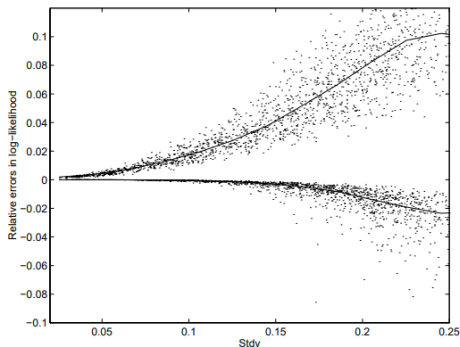
Log form.

$$\log P(\{S_i\}_{i \in L_1}) \geq \sum_{i \in L_1} \sum_{k=0}^{\infty} \left[-S_i \log\left(1 + \prod_{j \in L_2} (\mu_j e^{-2^k \lambda_{ij}} + (1 - \mu_j))\right) \right] \\ - \sum_{i \in L_1} (1 - S_i) \sum_{j \in L_2} \lambda_{ij} \mu_j + H(Q).$$

Experiments: sigmoid networks

Setup. Complete bipartite 8×8 ($L_2 \rightarrow L_1$). Weights $\theta_{ij} \sim \mathcal{N}(0, \sigma^2)$. Set $P(S_j = 1) = 1/2$ for all j .

Metric. Relative error in log-likelihood: $(\log P_{\text{bound}} / \log P) - 1$ vs. inter-layer dependence $\sigma_{\text{std}} = \sqrt{\text{Var}(P(S_i = 1 \mid \text{pa}[i]))}$.

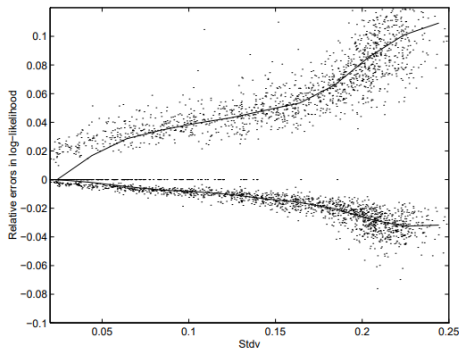


Solid lines - median relative error. Upper bound degrades faster as coupling increases. Tight when layers are weakly coupled.

Experiments: noisy-OR networks

Setup. Complete bipartite 8×8 ; parameters q_{ij} from a Dirichlet-controlled prior (Beta(1, n) with $q_{ij} \sim n(1 - q_{ij})^{n-1}$).

Metric. Same relative log-likelihood error vs. σ_{std} .



Solid lines - median relative error. Upper bound exact for $S_i = 0$. Lower bound a bit looser than sigmoid case. Upper bound deteriorates more quickly with stronger coupling.

Conclusion

- ▶ Variational upper bounds and lower bounds (generic) provide *rigorous*, efficiently computable likelihood bracketing in otherwise intractable BN settings.
- ▶ For both sigmoid and noisy-OR, the approach yields practically tight bounds in weak-to-moderate coupling regimes and degrades gracefully with stronger dependence.
- ▶ Enables confidence intervals and principled decision-making without exact inference.